Peruvian University of Applied Sciences

Faculty of Engineering

Computer Science Program

PC2

COURSE: Introduction to Deep Learning

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1. Introduction

Blood cell classification via microscopy is essential for diagnosing hematological conditions. However, manual analysis is resource-intensive. Convolutional neural networks have proven to be highly effective for medical image classification, yet developing practical models requires understanding how network architecture and reuse of pretrained models affect performance. The Blood Cell Image dataset from Kaggle is a public open-domain resource containing over 12,500 microscopic images classified into four blood cell types. Its direct clinical application makes it ideal for investigating these design questions. This project implements two classical architectures in original and modified forms, analyzes the impact of Batch Normalization on training dynamics, and evaluates Transfer Learning strategies to maximize accuracy with limited medical data.

2. Methodology

2.1 Preprocessing & EDA

We conducted exploratory data analysis to understand the dataset's statistical properties and determine necessary preprocessing steps. Through EDA, we identified valuable characteristics of the images, including resolution, channel distributions, and class balance.

Based on our analysis, we resized all images to 64×64 pixels for LeNet-5 and VGG-11 training from scratch. For Transfer Learning experiments, we used 224×224 pixels with ImageNet normalization statistics. Data augmentation was applied exclusively to the training split to prevent data leakage

2.2 Architecture Implementation

We implemented simplified versions of LeNet-5 and VGG-11 from scratch. For each architecture, we trained two variants: one baseline and one with Batch Normalization applied after each convolutional layer. This controlled comparison allows us to isolate the effect of normalization on training dynamics.

All 4 models were trained with the following hyperparameters:

- Optimizer: Adam
- Learning Rate: 1e-3
- Batch Size: 32
- Epochs: 20
- Random Seed: 42

2.3 Transfer Learning Experiment

We employed a pretrained model from ImageNet to investigate three Transfer Learning strategies:

1. Feature Extraction: Frozen convolutional layers with only the final fully connected layer trainable (Learning Rate of 1e-3).
2. Partial Fine-tuning: First two blocks frozen. Final blocks and FC layer trainable (Learning Rate of 1e-4)
3. Complete Fine-tuning: All layers unfrozen with a reduced learning rate (Learning rate of 1e-5)

The goal is to compare the performance of all three strategies against the best model trained from scratch (Section 2.2) to assess the value of Transfer Learning in a medical context.

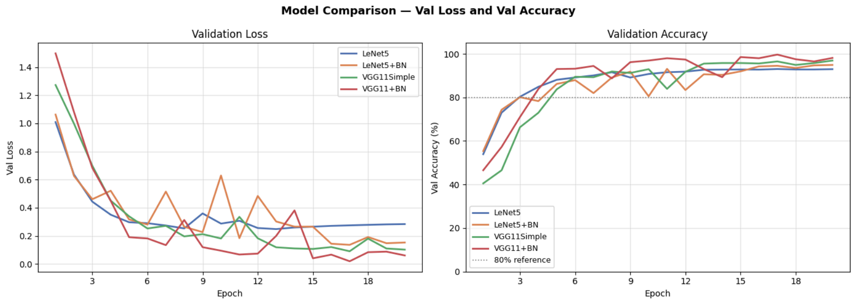
3. Results

The following table summarizes the comparison of all models. VGG-11 achieved the highest test accuracy among them at 89.06%, followed by VGG-11+BN, while both variants of LeNet-5 performed similarly to each other.

Model	Test Accuracy	Parameters	Avg. time/epoch
LeNet-5	63.57%	~338K	13.7s
LeNet-5 + BN	69.16%	~339K	12.3s
VGG-11 Simple	89.06%	~3.1M	12.4s
VGG-11 Simple + BN	87.41%	~3.1M	13.2s
ResNet-18 Feature Ext.	60.47%	~2K trainable	16.9s
ResNet-18 Partial FT	80.30%	~5.2M trainable	16s

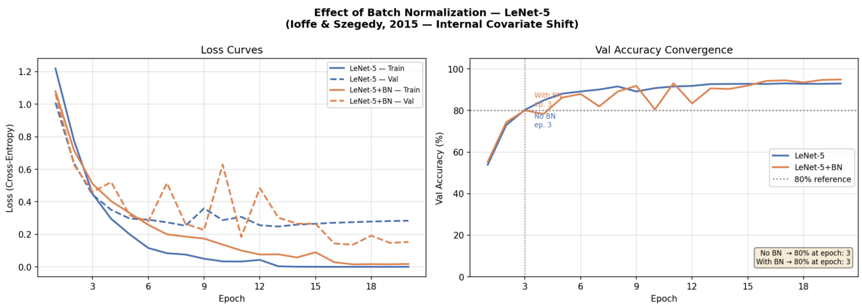
ResNet-18 Full FT	85.32%	~11.2M trainable	16.7s

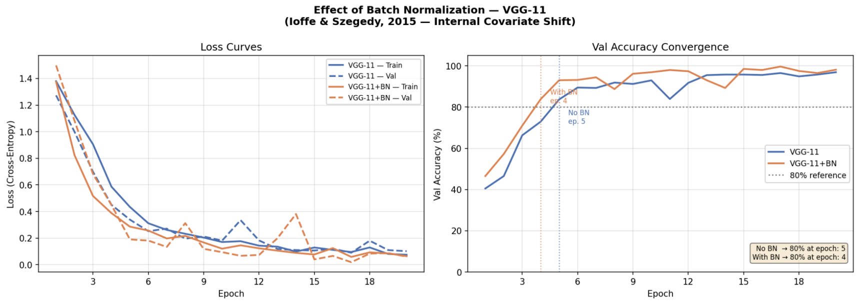
The model comparison plot below summarizes the validation loss and accuracy trajectories of the 4 models throughout training.



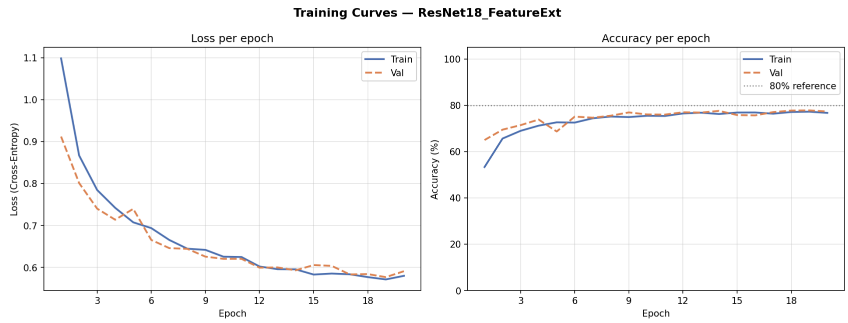
LeNet-5 variants reached 80% validation accuracy by epoch 3, while VGG-11 Simple reached it at epoch 5 and VGG-11+BN at epoch 4. LeNet-5 maintains the highest final validation loss, indicating limited model capacity. Both VGG-11 variants continue improving throughout training.

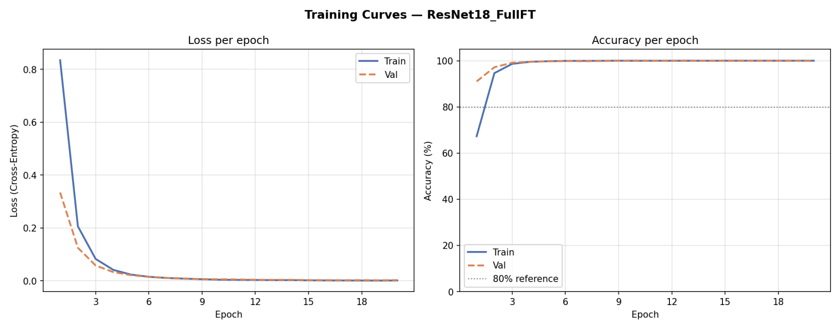
To isolate the effect of batch normalization, both architectures were trained under identical conditions with the only variable being the presence or absence of BN layers. For LeNet-5, both variants reached 80% validation accuracy at epoch 3. For VGG-11, BN reduced the epochs needed to reach 80% from epoch 5 to epoch 4, and produced lower and more stable validation loss throughout training.

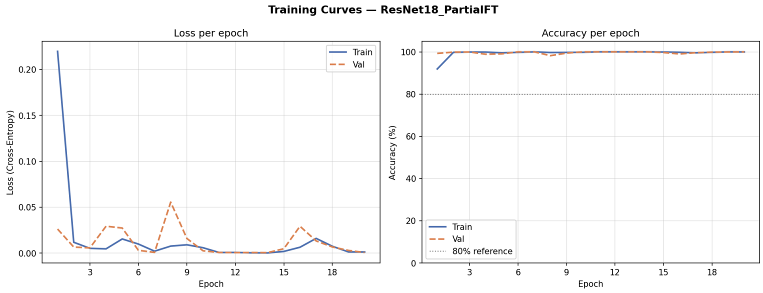




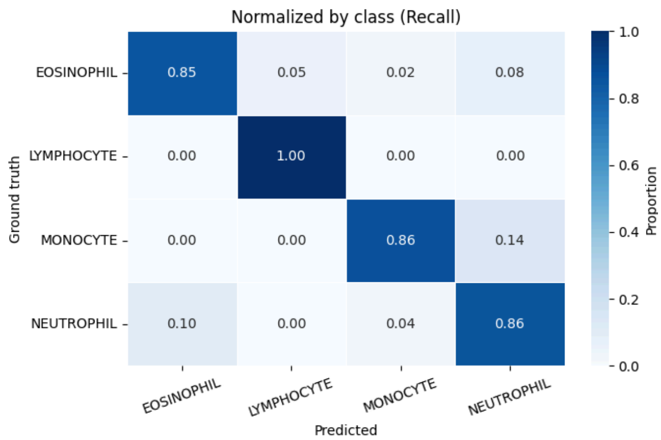
Additionally, 3 Transfer Learning strategies were evaluated using a pretrained ResNet-18. Feature extraction achieved only 60.47% test accuracy, never crossing the 80% validation accuracy threshold across all 20 epochs, with validation loss stabilizing around 0.60. Partial fine-tuning achieved near-perfect validation accuracy from epoch 2 onward, reaching a test accuracy of 80.30%. Full fine-tuning showed the smoothest convergence of all models, with both train and validation curves approaching 100% accuracy by epoch 5, achieving the highest TL test accuracy at 85.32%.







The confusion matrix for VGG-11 Simple shows more balanced per-class recall, with perfect LYMPHOCYTE recall and above 0.85 for the remaining classes.



4. Discussion

VGG-11 Simple outperformed LeNet-5 in both convergence speed and final test accuracy, consistent with Simonyan & Zisserman (2015), who demonstrated that deeper architectures with smaller convolutional filters achieve stronger feature extraction. LeNet-5, originally designed for grayscale document recognition on low-resolution inputs (LeCun et al., 1998), showed capacity limitations despite the input adaptation, with its training loss reaching near zero while validation loss stabilized around 0.28. This is a sign of overfitting due to the model's limited representational power relative to the task complexity.

Batch Normalization, introduced by Ioffe & Szegedy (2015) to reduce internal covariate shift, had notably different effects depending on architecture depth. For LeNet-5, both variants reached 80% validation accuracy at epoch 3, showing no convergence difference at this threshold. For VGG-11, BN reduced the epochs needed to reach 80% from epoch 5 to epoch 4, with the benefit being more pronounced for the deeper model. Regarding loss stability, BN variants produced noticeably smoother validation loss curves with fewer short-term oscillations, consistent with BN's mechanism of normalizing mini-batch activations and stabilizing gradient updates as described by Ioffe & Szegedy (2015). This effect was further confirmed in the high learning rate experiment (LR = 0.01), where the BN variant maintained stable training while the non-BN variant exhibited large oscillations and divergence under the same conditions. While BN consistently improved training stability, architecture depth remained the dominant factor since VGG-11 outperformed LeNet-5 regardless of BN, suggesting that capacity and depth cannot be substituted by normalization alone.

The three ResNet-18 Transfer Learning strategies produced different outcomes. ResNet-18 (He et al., 2016) was pretrained on ImageNet and applied to the blood cell dataset using three levels of adaptation. Feature extraction stabilized below 80% validation accuracy across all 20 epochs, with validation loss stabilizing around 0.60. This confirms that frozen ImageNet features lack the domain-specific higher-level representations needed to distinguish blood cell morphologies, a limitation consistent with the observations of Krizhevsky et al. (2012), who noted that deeper convolutional features are more task-specific and therefore more sensitive to domain shift.

Partial fine-tuning achieved near-perfect validation accuracy. Full fine-tuning showed similarly strong performance and has the most stable training behavior of all models tested. Its test accuracy of 85.32% is, however, lower than VGG-11 Simple (89.06%), and only marginally higher than partial fine-tuning (80.30%), despite training significantly more parameters.

Several limitations should be acknowledged. The unexpectedly lower test accuracy of ResNet-18 Full FT compared to VGG-11+BN warrants caution. The very small learning rate combined with 20 epochs may not have allowed adequate adaptation of early layers. Additionally, validation loss curves for LeNet-5+BN and VGG-11+BN show oscillations across epochs. Feature extraction never reaching 80% accuracy suggests that the visual domain gap between ImageNet and microscopic blood cell images is significant enough to require at least partial fine-tuning.

For future work, increasing the number of training epochs with early stopping based on test set performance could help close the gap between validation and test accuracy observed in the full fine-tuning strategy. For the models made from scratch, stronger data augmentation could reduce the overfitting visible in LeNet-5's train/val loss divergence.

5. Conclusions

VGG-11 Simple outperformed LeNet-5 in both convergence speed and final test accuracy. Furthermore, LeNet-5 training loss fell near zero while validation loss stabilized around 0.28, indicating overfitting and insufficient representational power for the blood-cell classification problem.

Among ResNet-18 transfer strategies, frozen feature extraction failed to exceed 80% validation accuracy, underscoring the domain gap between ImageNet features and microscopic blood-cell morphology. Partial and full fine-tuning produced strong results, with partial fine-tuning achieving the best validation performance and full fine-tuning showing the most stable training but a lower test accuracy than VGG-11 Simple, despite training significantly more parameters. Future work should extend training with early stopping or employing staged unfreezing for ResNet-18. Batch Normalization improved training stability and convergence speed, with its benefits being more pronounced in the deeper VGG-11 architecture. Finally, for medical imaging contexts with limited data, partial fine-tuning is the recommended strategy, as it achieves strong test accuracy with fewer trainable parameters and lower computational cost than full fine-tuning.

6. References

He, K., Zhang, X., Ren, S., & Sun, J. (2016, June 27-30). *Deep residual learning for image recognition* [Conference presentation]. 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Las Vegas, USA. https://doi.org/10.1109/CVPR.2016.90

Ioffe, S., & Szegedy, C. (2015). Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift. *Proceedings of Machine Learning Research*, *37*, 448-456. https://doi.org/10.48550/arXiv.1502.03167

Krizhevsky, A., Sutskever, I., & Hinton, G. (2012). ImageNet classification with deep convolutional neural networks. *Communications of the ACM*, *60*(6), 84–90. https://doi.org/10.1145/3065386

LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, *86*(11), 2278–2324. https://doi.org/10.1109/5.726791

Simonyan, K., & Zisserman, A. (2015). Very deep convolutional networks for large-scale image recognition. *Proceedings of the International Conference on Learning Representations (ICLR)*. https://doi.org/10.48550/arXiv.1409.1556